HACKAMATCH 1.0 / DATA SCIENCE + AI CHALLENGE

CityPulse

PREDICT. EXPLAIN. ACT.

Build a data-driven system that helps a city spot emerging problems, understand the signals behind them, and decide what to do next.

THE CHALLENGE Turn public urban data into a useful decision-support product.	**THE FREEDOM** Choose a city issue and build your own solution.
THE EXPECTATION Evidence-led analysis and honest evaluation.	**THE CONSTRAINT** Public data; no paid APIs or required paid services.

The strongest solution is not the one with the most complicated model. It is the one that turns reliable evidence into a decision a real person can understand and use.

01 / THE PROBLEM

Cities generate signals. Can you turn them into action?

Urban issues rarely exist in isolation. Congestion may coincide with rainfall. Air quality changes with weather and activity. Public complaints may reveal recurring hotspots. A useful system must do more than show charts: it should help people identify a meaningful pattern, estimate what may happen next, and decide where attention is needed.

Your challenge

Design and build **CityPulse**, a data-driven decision-support prototype focused on one clearly defined urban problem. Use public data to discover a pattern, quantify or forecast a risk where feasible, explain the evidence behind your output, and recommend a practical next step.

Choose one focused problem

- **Mobility:** congestion patterns, commute-time variation, or transport demand.
- **Environment:** air quality, heat, rainfall, or local environmental risk.
- **Public services:** complaint hotspots, service delays, or recurring infrastructure issues.
- **Urban resilience:** locations or periods that may need extra attention during adverse conditions.

These are example directions, not a required list. Teams may propose another urban issue if the data is public, the scope is realistic, and the outcome can be evaluated.

The core loop

DISCOVER Find a signal.	**PREDICT** Estimate what may happen next, if feasible.	**EXPLAIN** Show evidence and uncertainty.	**ACT** Recommend a next step.

Prediction is not mandatory when the dataset cannot support a valid forecast. A strong descriptive or risk-prioritisation system is acceptable if the team explains what additional data prediction would require.

02 / DATA & ACCESS

Public data. Reproducible work. Equal access.

The challenge must be feasible without a paid account, paid API, commercial dataset, or subscription. Teams should not gain an advantage by purchasing data or access to a model.

Suggested public starting points

SOURCE	POSSIBLE USE
India Open Government Data Platform https://www.data.gov.in/	Indian government datasets across multiple domains; check dataset metadata and licence.
OpenAQ https://openaq.org/	Open air-quality data and documentation; verify current access and coverage.
Central Pollution Control Board https://cpcb.nic.in/	Official Indian air-quality information and reports.
India Meteorological Department https://mausam.imd.gov.in/	Weather observations, forecasts, and climate information where available.
World Bank Open Data https://data.worldbank.org/	Public development indicators for broader urban context.
OpenStreetMap https://www.openstreetmap.org/	Geographic data for mapping; follow attribution and usage policies.

Mandatory data and tooling rules

- Use data that every team can access at no cost. Downloadable public files are encouraged when permitted by source terms.
- No paid APIs, paid model endpoints, paid datasets, subscriptions, or required credit-card trials for core functionality.
- Free APIs or free-tier services are acceptable only if the core submission remains reproducible without temporary promotional credits or exclusive access.
- Document dataset source URLs, access dates, licences/terms, fields used, filtering, missing-value handling, and transformations.
- Do not scrape or reuse data in violation of terms, access controls, or applicable law. Respect rate limits and attribution.
- Synthetic data may be used for clearly labelled UI testing, but not presented as real-world evidence.

Teams must verify current licence, terms, availability, and suitability. These links are starting points, not a guarantee that every desired variable is available.

03 / WHAT TO BUILD

Minimum viable solution

A complete submission should tell a connected story from raw data to a useful decision. A beautiful dashboard without sound analysis is incomplete; a strong model with no understandable user experience is also incomplete.

REQUIREMENT	WHAT TEAMS SHOULD SHOW	LEVEL
1. Problem framing	Define the issue, intended user, geographic/time scope, and decision supported.	**Core**
2. Data preparation	Explain sources, schema, missing values, duplicates, outliers, and transformations.	**Core**
3. Exploratory analysis	At least three useful findings supported by charts, summaries, or maps.	**Core**
4. Analytical method	Use a baseline, rule-based method, statistical analysis, or suitable ML approach; justify it.	**Core**
5. Evaluation	Validate with a holdout, chronological split, comparison, sensitivity analysis, or documented limitations.	**Core**
6. Decision interface	Show location/time, key signal, evidence, estimate where applicable, and a practical next step.	**Core**
7. Advanced extension	Try spatiotemporal modelling, uncertainty estimates, optimisation, anomaly detection, or scenario comparison.	**Stretch**

Example product flow

- A user selects a location, neighbourhood, time period, or urban issue.
- The interface presents the main pattern, a map/chart, and evidence supporting it.
- Where valid, the system shows a forecast or risk estimate and communicates uncertainty.
- The user sees a practical action or priority, plus assumptions and limitations.

Do not claim to predict future events unless the data and evaluation support it. A well-justified analysis can outperform an unnecessarily complex model.

04 / ANALYSIS & VALIDATION

Show your reasoning, not just your output.

There is no mandatory algorithm. Choose a method appropriate to the data and compare it with a simple baseline where possible.

Possible approaches

- **Exploratory analysis:** trends over time, geographic comparisons, correlations, distributions, and outlier analysis.
- **Forecasting:** compare with a naive or seasonal baseline; use chronological validation when predicting the future.
- **Anomaly detection:** define what counts as unusual and check whether flagged examples are plausible.
- **Risk scoring/classification:** define the target and evaluate precision/recall or other relevant metrics when labels exist.
- **Clustering:** explain what groups represent and whether they are useful for the intended decision.
- **Resource prioritisation:** make constraints explicit and show how recommendations change under different assumptions.

Evaluation principles

- Use a baseline and report the test/validation method.
- Avoid data leakage: information unavailable at decision time must not enter model features.
- Report missing labels, uneven coverage, sampling bias, and differences between locations or time periods.
- Use appropriate metrics: MAE/RMSE for numeric forecasts; precision/recall for labelled detection tasks; calibration checks for probability estimates.
- Do not treat correlation as proof of causation. Keep recommendations proportional to the evidence.
- If reliable ground truth is unavailable, state that limitation and evaluate what can be measured instead.

A strong submission answers: What did we observe? How reliable is it? What might happen next? What action could help—and what could make our conclusion wrong?

Responsible use

CityPulse is a prototype, not an official emergency-warning system or a substitute for expert judgement. Avoid unsupported claims, expose uncertainty where relevant, and consider how errors could affect people or neighbourhoods. Do not publish sensitive personal data.

05 / SUBMISSION PACKAGE

What every team must submit

Working prototype. A runnable web app, dashboard, map, notebook-based interface, or clearly documented interactive demo.

Source code. Repository or source archive with setup instructions and dependency list.

Data statement. Dataset links, access dates, licence/terms, fields used, and preprocessing steps.

Analysis and method. Notebook or report covering EDA, baseline/method, evaluation, and limitations.

Presentation. Concise slides explaining problem, insight, approach, results, product, and next steps.

Reproducibility notes. How judges can run the core demo without paid services, secret keys, or private datasets.

Suggested 5-minute finalist presentation

TIME	CONTENT
0:00–0:30	Problem, target user, and why it matters.
0:30–1:15	Dataset, data quality, and strongest analytical insight.
1:15–2:00	Method, baseline, and validation approach.
2:00–4:30	Live product demo: signal → explanation → action.
4:30–5:00	Limitations, next steps, and evidence needed to increase confidence.

Teams should prepare backup screenshots or a recording in case the live demo fails. A backup supports continuity but does not replace a working, reproducible submission.

06 / JUDGING

How submissions will be evaluated

Eligible submissions will be assessed against the same rubric. Judges should reward clear reasoning, sound analysis, practical usefulness, and reproducibility—not complexity for its own sake.

CRITERION	POINTS
Problem framing and relevance	**10**
Data sourcing, cleaning, and EDA	**20**
Method selection and analytical/model quality	**15**
Evaluation, validation, and integrity	**15**
UI/UX and decision usefulness	**20**
Explainability, uncertainty, and limitations	**10**
Implementation and reproducibility	**5**
Presentation and completeness	**5**
TOTAL	**100**

Top-10 shortlist recommendation

- First verify eligibility and completeness against the published rules.
- Have at least two judges score each eligible submission independently; average their scores.
- Rank by average score. For ties, prioritise evaluation integrity, then data analysis, then UI/UX.
- Shortlisted teams present for 5 minutes, followed by 3 minutes of judge Q&A.; Apply scoring consistently and avoid double-counting criteria already scored.

Disqualification must follow published rules consistently. Do not reject teams solely for choosing a different valid urban problem or a simpler appropriate method.

07 / RULES & EXPECTATIONS

Build honestly. Credit clearly. Make it reproducible.

Eligibility and scope

- Teams may choose one urban problem and define a realistic scope that can be demonstrated during the hackathon.
- A working core solution matters more than an unfinished collection of features.

Data, tools, and AI assistance

- All core datasets and services must be publicly accessible at no cost. No paid APIs, paid datasets, paid model endpoints, or paid subscriptions may be required to run the submission.
- Open-source libraries and public models are allowed subject to their licences. Disclose major third-party components and AI assistance.
- Do not include API keys, passwords, private credentials, or personal information in a public repository.

Originality and integrity

- Teams must be able to explain their implementation and submitted results.
- Do not fabricate data, metrics, citations, model outputs, or demo results. Label simulated/synthetic data explicitly.
- Do not copy another team's work or misrepresent third-party code as original. Cite adapted work.
- Respect dataset licences, source terms, privacy, and applicable law.

Submission and judging

- Follow the official deadline, format, team-size rules, and presentation schedule announced by organisers.
- Judges may ask teams to explain their pipeline, validation method, assumptions, and how to reproduce the demo.
- Organisers may disqualify material rule violations, fabricated evidence, plagiarism, or failure to meet the free-access requirement, applying rules consistently.

Ownership and publicity

Teams retain ownership of original work, subject to third-party licences and event terms communicated by organisers. Teams may be asked for permission before their work is publicly showcased.

CITYPULSE

Predict. Explain. Act.

A city is full of data. Your job is to turn the right signals into better decisions.

Organised by Data Mavericks × NSDC | HackAMatch 1.0